

PROBABILISTIC SYSTEMS ANALYSIS AND APPLIED PROBABILITY

MIHIR RAO

Contents

1	Probability Models and Axioms	2
1.1	Sample spaces: what can happen?	2
1.2	Events and probability axioms	3
1.3	A finite probability law: two die rolls	4
1.4	The discrete uniform law	5
1.5	The continuous uniform law	5
1.6	A countably infinite law and countable additivity	6
2	Conditioning and Bayes' Rule	8
2.1	Start with a probability model	8
2.2	Conditional probability: restrict and renormalize	8
2.3	Building a model from conditional probabilities	10
2.4	The multiplication rule	11
2.5	Total probability: combine disjoint cases	12
2.6	Bayes' rule: reverse the conditioning	13

LECTURE 1: PROBABILITY MODELS AND AXIOMS

Probability gives us a mathematical way to reason about uncertainty. A **PROBABILITY MODEL** specifies what can happen and assigns probabilities to events. Its two ingredients are a sample space and a probability law.

1.1 Sample spaces: what can happen?

Before assigning probabilities, decide what one complete outcome records. The detail must be enough to answer the questions of interest.

Definition 1.1 [SAMPLE SPACE]. The **SAMPLE SPACE**, denoted by Ω , is the set of all possible outcomes of an experiment. Its outcomes must be **MUTUALLY EXCLUSIVE**: only one outcome occurs in a run. They must also be **COLLECTIVELY EXHAUSTIVE**: every possible result is included.

Example: *Two rolls of a tetrahedral die.* Each roll produces one of 1, 2, 3, 4. Write X for the first result and Y for the second. One complete outcome is the ordered pair (X, Y) , so

$$\Omega = \{(x, y) : x, y \in \{1, 2, 3, 4\}\}.$$

There are $4 \times 4 = 16$ outcomes. The order matters: $(1, 2)$ and $(2, 1)$ describe different results. We can display the pairs in a grid or describe the two rolls sequentially with a tree. Each full path through the tree ends at one of the same 16 outcomes.

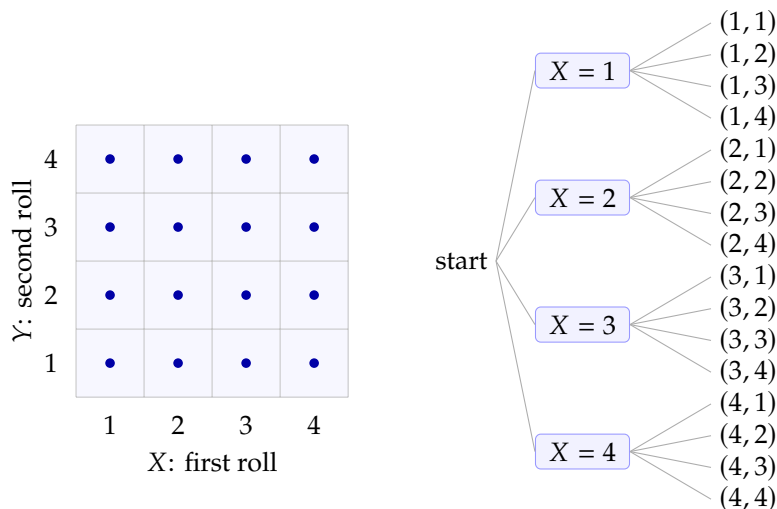


Figure 1: The grid and the tree describe the same sample space. A cell or a complete path records both rolls.

Example: *Two numbers in the unit interval.* If each result can be any real number from 0 to 1, the sample space is

$$\Omega = \{(x, y) : 0 \leq x \leq 1, 0 \leq y \leq 1\} = [0, 1]^2.$$

It is the entire unit square, including its boundary. The die space is finite; this space is continuous. Both still need a probability law.

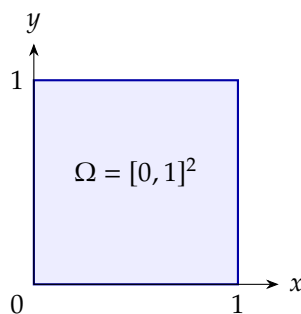


Figure 2: Each point in the unit square is one possible ordered pair.

1.2 Events and probability axioms

We often ask whether a result satisfies a condition, rather than whether one particular outcome occurs.

Definition 1.2 [EVENT AND PROBABILITY LAW]. An event A is a subset of Ω : it occurs when the outcome belongs to A . A PROBABILITY LAW assigns a number $P(A)$ to each event in its domain, subject to the probability axioms below.

For a finite or countably infinite sample space, every subset can be an event. On a continuous space, the domain of the law must be chosen with care; we use ordinary geometric regions here.

Definition 1.3 [PROBABILITY AXIOMS]. For events in the domain of P :

1. NONNEGATIVITY: $P(A) \geq 0$ for every event A .
2. NORMALIZATION: $P(\Omega) = 1$.
3. ADDITIVITY: for DISJOINT events A and B , meaning $A \cap B = \emptyset$,

$$P(A \cup B) = P(A) + P(B).$$

The first two axioms say that probabilities cannot be negative and that some outcome in the sample space is certain to occur. The third lets us add probabilities when the events cannot occur together. For overlapping events, this sum would count the common outcomes twice.

Distinct individual outcomes give disjoint singleton events. Repeatedly applying additivity therefore gives

$$P(\{s_1, \dots, s_k\}) = P(\{s_1\}) + \dots + P(\{s_k\}).$$

We may abbreviate $P(\{s\})$ to $P(s)$. The two-event axiom handles finite unions; an infinite union will require a stronger form of additivity.

1.3 A finite probability law: two die rolls

Return to the 16 ordered die outcomes and assign probability $1/16$ to each. The probabilities are nonnegative and sum to $16(1/16) = 1$. An event containing k outcomes has probability $k/16$.

Example: *Either (1, 1) or (1, 2).* These are two distinct outcomes, so

$$P(\{(1, 1), (1, 2)\}) = \frac{1}{16} + \frac{1}{16} = \frac{1}{8}.$$

Example: *The first roll is 1.* The second roll can be 1, 2, 3, or 4. Thus

$$\{X = 1\} = \{(1, 1), (1, 2), (1, 3), (1, 4)\}, \quad P(X = 1) = \frac{4}{16} = \frac{1}{4}.$$

The notation $\{X = 1\}$ names the subset of outcomes satisfying the condition.

Example: *The sum is odd.* One roll must be odd and the other even. There are two odd values and two even values, so there are $2 \times 2 = 4$ pairs of each order. Hence

$$P(X + Y \text{ is odd}) = \frac{4 + 4}{16} = \frac{1}{2}.$$

Example: *The smaller roll is 2.* Both rolls must be at least 2, and at least one must equal 2. The event is

$$\{\min(X, Y) = 2\} = \{(2, 2), (2, 3), (2, 4), (3, 2), (4, 2)\}.$$

There are five outcomes, giving

$$P(\min(X, Y) = 2) = \frac{5}{16}.$$

Counting three pairs with $X = 2$ and three with $Y = 2$ would count $(2, 2)$ twice, so it must be included only once.

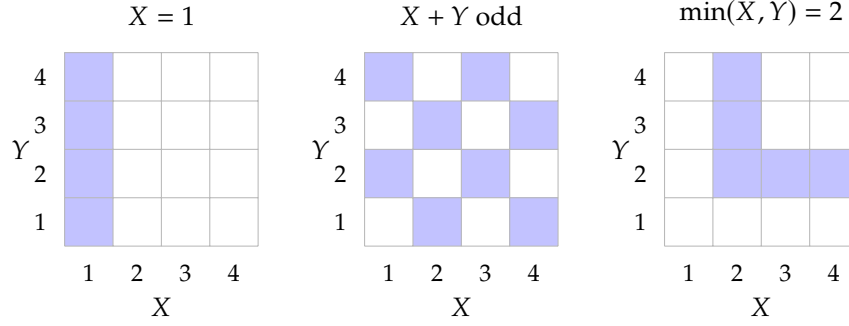


Figure 3: Highlighted cells show the last three die events. Each cell has probability $1/16$. In the last panel, the corner $(2, 2)$ is counted once.

1.4 The discrete uniform law

When a finite sample space has N equally likely outcomes, normalization forces each outcome to have probability $1/N$. This is the **DISCRETE UNIFORM LAW**. Additivity then gives

$$P(A) = \frac{|A|}{|\Omega|},$$

where $|A|$ is the number of outcomes in A and $|\Omega| = N$. Computing a probability becomes a counting problem. Fair coins, fair dice, and well-shuffled decks are modeled this way. The formula relies on the outcomes being equally likely.

1.5 The continuous uniform law

Now give the unit square a **CONTINUOUS UNIFORM LAW**: the probability of a region is its area. The square's total area is 1, so the law is normalized. Equal-area regions have equal probabilities, and areas add for disjoint regions.

Example: *The two numbers sum to at most $1/2$.* The event is

$$A = \{(x, y) \in [0, 1]^2 : x + y \leq \frac{1}{2}\}.$$

It is the triangle bounded by the two coordinate axes and the line $x + y = 1/2$. Both perpendicular sides have length $1/2$, so

$$P(X + Y \leq \frac{1}{2}) = \text{area}(A) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}.$$

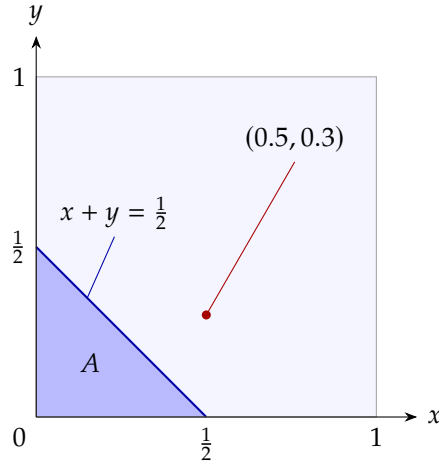


Figure 4: The shaded triangle has area $1/8$. The point $(0.5, 0.3)$ is a possible outcome, but its singleton has zero area.

Example: *The exact pair* $(0.5, 0.3)$. A single point has zero area, so

$$P((X, Y) = (0.5, 0.3)) = 0.$$

The point is still in the sample space. A probability-zero event need not be impossible: in this model every individual point has probability zero, while the whole square has probability one.

1.6 A countably infinite law and countable additivity

Consider the sample space $\Omega = \{1, 2, 3, \dots\}$ with

$$P(n) = 2^{-n}, \quad n = 1, 2, \dots$$

Each probability is positive, and the geometric series checks normalization:

$$\sum_{n=1}^{\infty} 2^{-n} = \frac{1/2}{1 - 1/2} = 1.$$

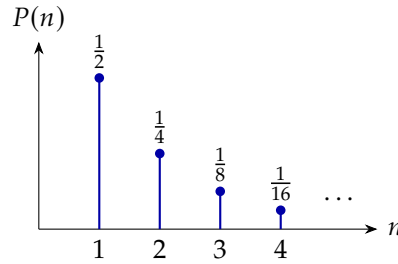


Figure 5: The first four outcome probabilities. Each is half the preceding one, and the pattern continues for every positive integer.

Example: *The outcome is even.* The event $\{2, 4, 6, \dots\}$ contains infinitely many disjoint singleton events. Adding their probabilities gives

$$\begin{aligned} P(\{2, 4, 6, \dots\}) &= \sum_{k=1}^{\infty} 2^{-2k} = \frac{1}{4} + \frac{1}{16} + \frac{1}{64} + \dots \\ &= \frac{1/4}{1 - 1/4} = \frac{1}{3}. \end{aligned}$$

This calculation uses an infinite sum, which finite additivity alone does not justify.

Definition 1.4 [COUNTABLE ADDITIVITY]. If $A_1, A_2, \dots$ are pairwise disjoint events, the COUNTABLE ADDITIVITY axiom requires

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

This is the full additivity axiom for a probability law. It covers a sequence indexed by the positive integers and includes finite additivity as a special case. It justifies the even-outcome calculation above. It does not permit summing singleton probabilities over all the points of the unit square: that collection is uncountable.

Revision. A model needs both a sample space and a probability law. Events are sets of outcomes. Count outcomes under a finite uniform law, measure area under the uniform law on the unit square, and use countable additivity for a countable union of disjoint events.

Common mistakes: using the counting formula without equal likelihood, counting a shared outcome twice, or treating probability zero as impossibility.

Check yourself: Why do two tetrahedral-die rolls give 16 outcomes? Why does $\min(X, Y) = 2$ select five of them? Why can an infinite sum give the probability of an even integer, but cannot add up the probabilities of every point in the unit square?

Source. John Tsitsiklis, MIT 6.041/6.431, *Probabilistic Systems Analysis and Applied Probability*, Fall 2010, [Lecture 1: Probability Models and Axioms](#), PDF pp. 1–3. The calculations are completed and the diagrams redrawn here. Adapted from MIT OpenCourseWare under [CC BY-NC-SA 4.0](#).

LECTURE 2: CONDITIONING AND BAYES' RULE

Learning that an event occurred changes which outcomes remain possible and how their probabilities should be compared. Conditioning makes this update precise. It also gives three useful tools: multiplying probabilities along a sequence, combining disjoint cases, and reversing a conditional probability.

2.1 Start with a probability model

The sample space Ω lists mutually exclusive, collectively exhaustive outcomes at the detail needed for the question. Events are subsets of Ω . Their probabilities are nonnegative, satisfy $P(\Omega) = 1$, and add over disjoint events. For a disjoint sequence $A_1, A_2, \dots$,

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i).$$

To solve a problem, specify the sample space, define its probability law, identify the event of interest, and then calculate. A calculation is only meaningful once these ingredients are clear.

2.2 Conditional probability: restrict and renormalize

Suppose we learn that B occurred. Outcomes outside B can be discarded. Among the outcomes in B , the event A occurs exactly on $A \cap B$. We compare the probability of this intersection with the probability of the new universe B .

Definition 2.1 [CONDITIONAL PROBABILITY]. For events A and B with $P(B) > 0$, the **CONDITIONAL PROBABILITY** of A given B is

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

This definition does not assign a value when $P(B) = 0$.

Restricting to B keeps only the outcomes compatible with the information. Dividing by $P(B)$ rescales the remaining probability to one: $P(B \mid B) = P(B)/P(B) = 1$.

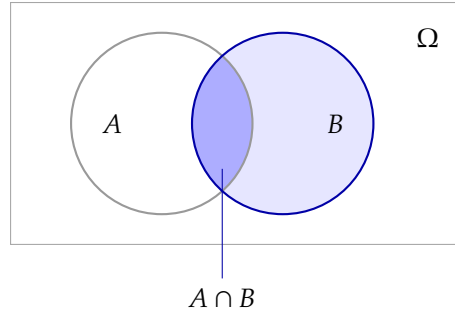


Figure 6: Conditioning keeps B and compares $A \cap B$ with all of B . This is a schematic set diagram; region sizes do not represent probabilities.

Example: *The minimum of two die rolls is 2.* Let X and Y be the two results of a tetrahedral die, with each of the 16 ordered pairs assigned probability $1/16$. Suppose we know

$$B = \{\min(X, Y) = 2\} = \{(2, 2), (2, 3), (2, 4), (3, 2), (4, 2)\}.$$

Then $P(B) = 5/16$. Each remaining pair has conditional probability $(1/16)/(5/16) = 1/5$.

Let $M = \max(X, Y)$. The event $M = 1$ has no outcome in common with B : if the minimum is 2, the maximum cannot be 1. Therefore

$$P(M = 1 \mid B) = \frac{0}{5/16} = 0.$$

For $M = 2$, the only compatible pair is $(2, 2)$, giving

$$P(M = 2 \mid B) = \frac{1/16}{5/16} = \frac{1}{5}.$$

The denominator counts the five outcomes allowed by the information, not all 16 outcomes in the original space.

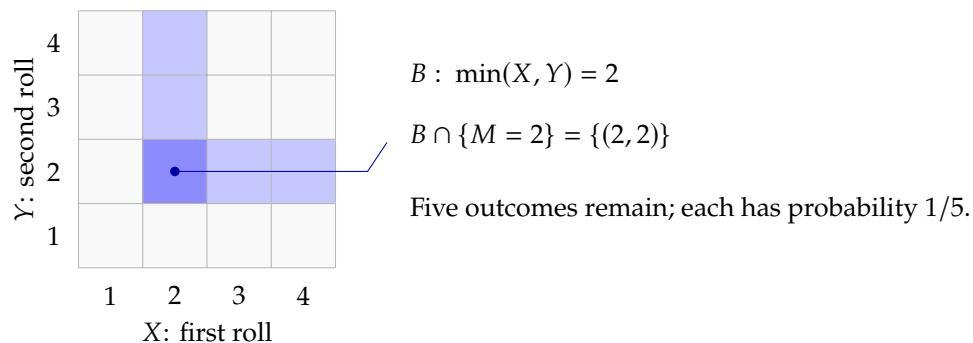


Figure 7: The five colored cells form B . The darker cell $(2, 2)$ is the only one where $M = 2$. Each colored cell has probability $1/5$ given B .

2.3 Building a model from conditional probabilities

A model can be specified in stages. First give the probability of one event, then give the probabilities of later events under each possible earlier result. Write A^c for the COMPLEMENT of A , the event that A does not occur.

Example: *An airplane and a radar reading.* Let A mean that an airplane is flying above, and let B mean that something registers on the radar screen. The model is

$$P(A) = 0.05, \quad P(A^c) = 0.95,$$

with the following conditional probabilities:

$$\begin{aligned} P(B | A) &= 0.99 & P(B^c | A) &= 0.01, \\ P(B | A^c) &= 0.10 & P(B^c | A^c) &= 0.90. \end{aligned}$$

Thus a registration is likely when a plane is present, but registrations also occur when no plane is present. Within each condition, the two radar probabilities sum to one.

Rearranging the conditional-probability definition gives

$$P(A \cap B) = P(A)P(B | A).$$

Multiply the probability of the first stage by the conditional probability of the second stage to obtain the probability of a complete path.

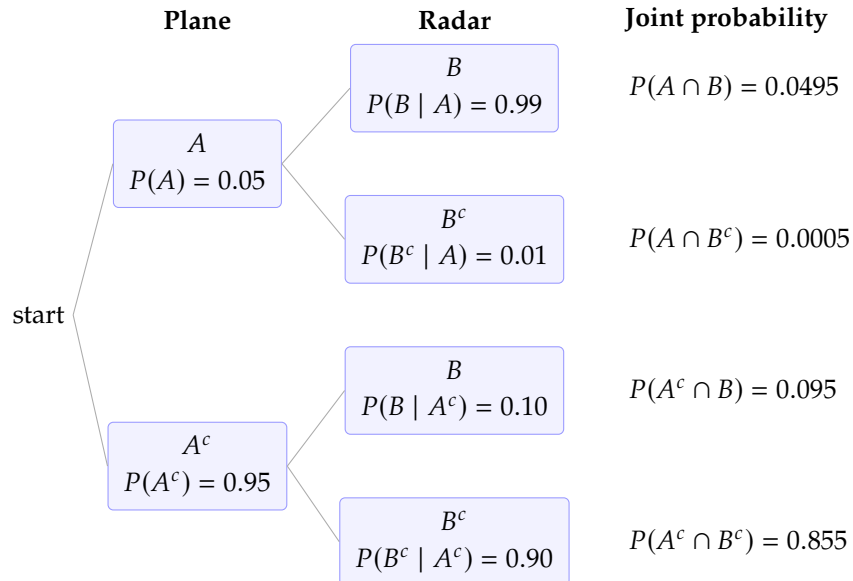


Figure 8: Each path specifies both plane presence and radar status. Multiply the two factors on a path to obtain the joint probability on the right.

For a plane and a registration,

$$P(A \cap B) = 0.05(0.99) = 0.0495.$$

A registration can come from either $A \cap B$ or $A^c \cap B$. These events are disjoint, so

$$\begin{aligned} P(B) &= P(A \cap B) + P(A^c \cap B) \\ &= 0.05(0.99) + 0.95(0.10) = 0.1445. \end{aligned}$$

Once a registration is observed, the probability of a plane is

$$P(A | B) = \frac{P(A \cap B)}{P(B)} = \frac{0.0495}{0.1445} = \frac{99}{289} \approx 0.3426.$$

The two directions answer different questions: $P(B | A) = 0.99$ describes the radar given a plane; $P(A | B) \approx 0.3426$ describes plane presence given a registration. The latter must account for how often planes are present and how often the radar registers without one.

2.4 The multiplication rule

For three events, a complete path specifies A , then B given A , then C given both earlier events. Each factor must use all the information accumulated so far.

Proposition 2.2 [MULTIPLICATION RULE]. *If $P(A) > 0$ and $P(A \cap B) > 0$, then*

$$P(A \cap B \cap C) = P(A)P(B | A)P(C | A \cap B).$$

The positivity assumptions ensure that the conditional probabilities on the right are defined.

Proof. Apply the conditional-probability definition twice: first condition on the whole event $A \cap B$, then express $P(A \cap B)$ by conditioning on A .

$$\begin{aligned} P(A \cap B \cap C) &= P(A \cap B)P(C | A \cap B) \\ &= P(A)P(B | A)P(C | A \cap B). \end{aligned}$$

□

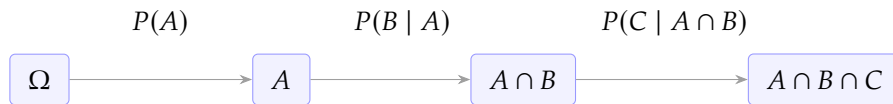


Figure 9: One path through a three-stage model. The last factor conditions on $A \cap B$, because both earlier events have been specified.

2.5 Total probability: combine disjoint cases

Sometimes the probability of B is easier to calculate after dividing the sample space into cases. Compute the contribution from each case, then add the contributions.

Definition 2.3 [PARTITION]. Events A_1, A_2, A_3 form a PARTITION of Ω if they are pairwise disjoint and cover the whole space:

$$A_i \cap A_j = \emptyset \quad (i \neq j), \quad A_1 \cup A_2 \cup A_3 = \Omega.$$

Every outcome belongs to exactly one of the three cases.

Theorem 2.4 [TOTAL PROBABILITY]. For a partition A_1, A_2, A_3 with $P(A_i) > 0$ for each i ,

$$P(B) = P(A_1)P(B \mid A_1) + P(A_2)P(B \mid A_2) + P(A_3)P(B \mid A_3).$$

Proof. Each outcome of B belongs to exactly one part of the partition, so $B = (A_1 \cap B) \cup (A_2 \cap B) \cup (A_3 \cap B)$ is a disjoint union. Additivity gives

$$P(B) = P(A_1 \cap B) + P(A_2 \cap B) + P(A_3 \cap B).$$

For each piece, use $P(A_i \cap B) = P(A_i)P(B \mid A_i)$. Substituting these three products gives the formula. $\square$

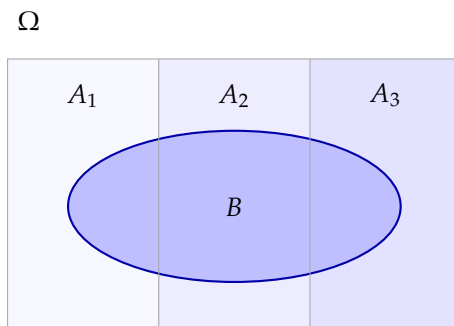


Figure 10: The partition cuts B into three disjoint pieces. Their probabilities add to $P(B)$. The geometry is schematic.

Each term weights a conditional probability by the probability of its case. In the radar example, the two cases are A and A^c , giving the calculation $P(B) = 0.05(0.99) + 0.95(0.10)$.

2.6 Bayes' rule: reverse the conditioning

Suppose we know how likely B is under each case A_i , but observe B and want to infer which case occurred. The PRIOR PROBABILITIES $P(A_i)$ describe the cases before using this observation. The updated probabilities are $P(A_i | B)$.

Theorem 2.5 [BAYES' RULE]. For the partition above, if $P(A_i) > 0$ for each i and $P(B) > 0$, then

$$P(A_i | B) = \frac{P(A_i)P(B | A_i)}{\sum_{j=1}^3 P(A_j)P(B | A_j)}, \quad i = 1, 2, 3.$$

Proof. The conditional-probability definition gives

$$P(A_i | B) = \frac{P(A_i \cap B)}{P(B)}.$$

The multiplication rule supplies the numerator, $P(A_i \cap B) = P(A_i)P(B | A_i)$. The total probability theorem supplies the denominator by adding the contributions from all three cases. Substituting both expressions yields Bayes' rule. $\square$

In [figure 10](#), the numerator is the probability of the piece $A_i \cap B$ and the denominator is the probability of all of B . Thus Bayes' rule compares one case's contribution with the total probability of the observation. In the radar model, it updates $P(A) = 0.05$ to $P(A | B) \approx 0.3426$ after a registration.

Revision. Conditioning restricts the possible outcomes and divides by the probability of the conditioning event. Multiply conditional factors along a path; add disjoint paths for total probability. Bayes' rule divides one case's contribution by the total probability of the observed event.

Common mistakes: interchanging $P(A | B)$ and $P(B | A)$, using a denominator that ignores the information, or forgetting earlier events in the last factor of the multiplication rule.

Check yourself: Why is $P(M = 2 | \min(X, Y) = 2) = 1/5$? Why is the radar model's $P(B) = 0.1445$? Which two ingredients, besides $P(B | A)$, are needed to calculate $P(A | B)$?

Source. John Tsitsiklis, MIT 6.041/6.431, *Probabilistic Systems Analysis and Applied Probability*, Fall 2010, [Lecture 2: Conditioning and Bayes' Rule](#), PDF pp. 1–2. The supplied calculations and derivations are completed, and the diagrams redrawn here. Adapted from MIT OpenCourseWare under [CC BY-NC-SA 4.0](#).